
PH102: Physics-2

Electrodynamics

Lecture # 14
Revision
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Representing source and field point

A source point, $\mathbf{r}'$, where an electric charge is located.

A field point, $\mathbf{r}$, at which the electric field is calculated.

A short-hand notation for the separation vector from the source point to the field point is

$$\mathbf{z} \equiv \mathbf{r} - \mathbf{r}'.$$

Its magnitude is

$$z = |\mathbf{r} - \mathbf{r}'|,$$

and a unit vector in the direction from $\mathbf{r}'$ to $\mathbf{r}$ is

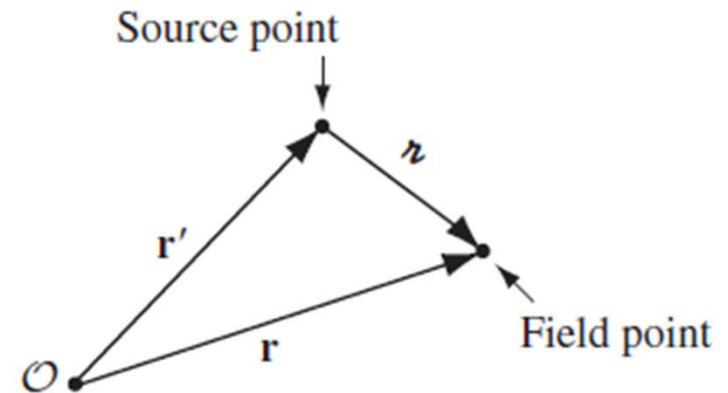
$$\hat{\mathbf{z}} = \frac{\mathbf{z}}{z} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}.$$

In Cartesian coordinates,

$$\mathbf{z} = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}},$$

$$z = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2},$$

$$\hat{\mathbf{z}} = \frac{(x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}$$



Gradient

$$\vec{\nabla} T = \left(\hat{\mathbf{i}} \frac{\partial T}{\partial x} + \hat{\mathbf{j}} \frac{\partial T}{\partial y} + \hat{\mathbf{k}} \frac{\partial T}{\partial z} \right)$$

The gradient has the formal appearance of a vector, ∇ , “multiplying”, a scalar T .

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}.$$

∇ is a **vector operator** that acts upon T and not a vector that multiplies T .

The Divergence

Divergence of a vector $\mathbf{v}$ is:

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) \cdot (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) \\ &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}. \end{aligned}$$

$\nabla \cdot \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ spread out from the point in question.
(divergence of a scalar is meaningless).

The curl

The curl of a vector is

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_x & v_y & v_z \end{vmatrix} \\ = \hat{\mathbf{x}} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right).$$

$\nabla \times \mathbf{v}$ is a measure of how much the vector $\mathbf{v}$ curl around the point in question.

Summary

- ✓ The notations **A**, **B**, **C** etc. are used for usual vectors which represent a meaningful physical quantity called a vector field.
- ✓ del (∇) is an operator and is not a usual vector.
- ✓ The operator operates on a scalar or a vector field and generates either a scalar or a vector field depending upon a valid operation of the form ∇T , $\nabla \cdot \mathbf{v}$ or $\nabla \times \mathbf{v}$.
- ✓ The vector identities need not necessarily be always true for the operators.

Fundamental theorems of calculus

The **fundamental theorem for gradients**; it says that the integral (here a *line* integral) of a derivative (here the *gradient*) is given by the value of the function at the boundaries (**a** and **b**).

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}).$$

The fundamental theorem for divergences states that: The integration of a derivative (the divergence) over a region (a volume) is equal to the value of the function at the boundary (the surface that bounds the volume).

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}.$$

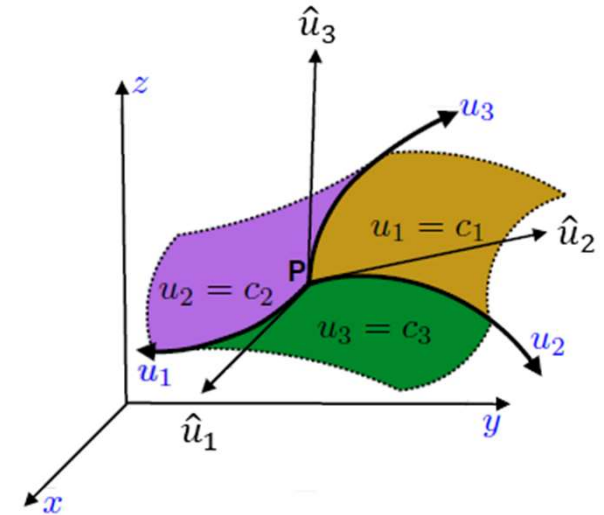
The fundamental theorem for curls (**Stokes' theorem**) states that: The integration of a derivative (here, the curl) over a region (**a patch of surface S**) is equal to the value of the function at the boundary (**the perimeter of the patch P**).

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_P \mathbf{v} \cdot d\mathbf{l}.$$

Generalized orthogonal curvilinear coordinates

The gradient, divergence and curl in terms of generalized curvilinear coordinates (u_1, u_2, u_3) are given by

$$\nabla T = \hat{u}_1 \frac{1}{h_1} \frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2} \frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{h_3} \frac{\partial T}{\partial u_3}$$



$$\nabla \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) + \frac{\partial}{\partial u_2} (v_2 h_3 h_1) + \frac{\partial}{\partial u_3} (v_3 h_1 h_2) \right]$$

$$\nabla \times \mathbf{v} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix}$$

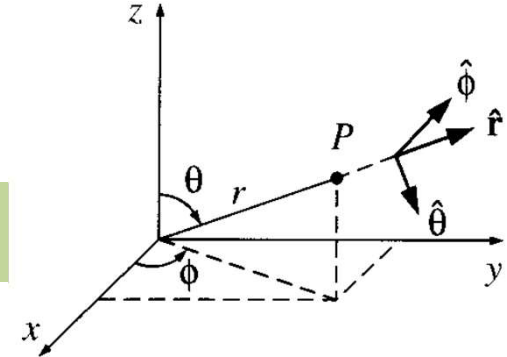
Orthogonal curvilinear polar coordinates

➤ Spherical polar coordinates

$$x = r \cos \theta \sin \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

$$d\mathbf{l} = \hat{\mathbf{r}}dl_r + \hat{\boldsymbol{\theta}}dl_\theta + \hat{\boldsymbol{\phi}}dl_\phi = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}}$$

$$h_1 = 1, h_2 = r, h_3 = r \sin\theta$$

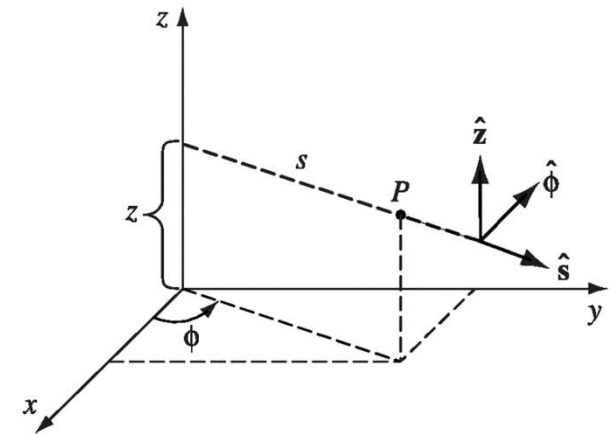


➤ Cylindrical polar coordinates

$$x = s \cos \phi, \quad y = s \sin \phi, \quad z = z$$

$$d\mathbf{l} = \hat{\mathbf{s}}dl_s + \hat{\boldsymbol{\phi}}dl_\phi + \hat{\mathbf{z}}dl_z = ds\hat{\mathbf{s}} + s d\phi\hat{\boldsymbol{\phi}} + dz\hat{\mathbf{z}}$$

$$h_1 = 1, h_2 = s, h_3 = 1$$



Dirac delta function

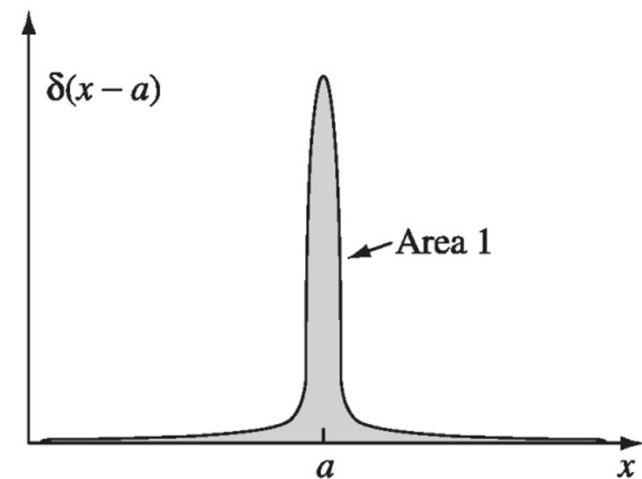
The term “**Delta function**” is a misnomer. Its value is not finite at $x = 0$. One can, however, think of delta function as the limit of a sequence of a function. In the mathematical literature it is known as a **generalized function**, or a **distribution**.

$$\delta(x - a) = \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases}, \quad \int_{-\infty}^{\infty} \delta(x - a) dx = 1$$

$$f(x)\delta(x - a) = f(a)\delta(x - a)$$

$$\int_{-\infty}^{\infty} f(x)\delta(x - a) dx = f(a)$$

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$



Gauss's law in differential form

❖ Continuous charge distribution

The electric field of a line charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\vec{r}')}{r^2} \hat{r} d\ell'$

The electric field of a surface charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\vec{r}')}{r^2} \hat{r} da'$

The electric field of a volume charge: $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{r^2} \hat{r} d\tau'$

❖ Flux through any surface that encloses bunch of charges $= \frac{Q_{\text{enc}}}{\epsilon_0}$

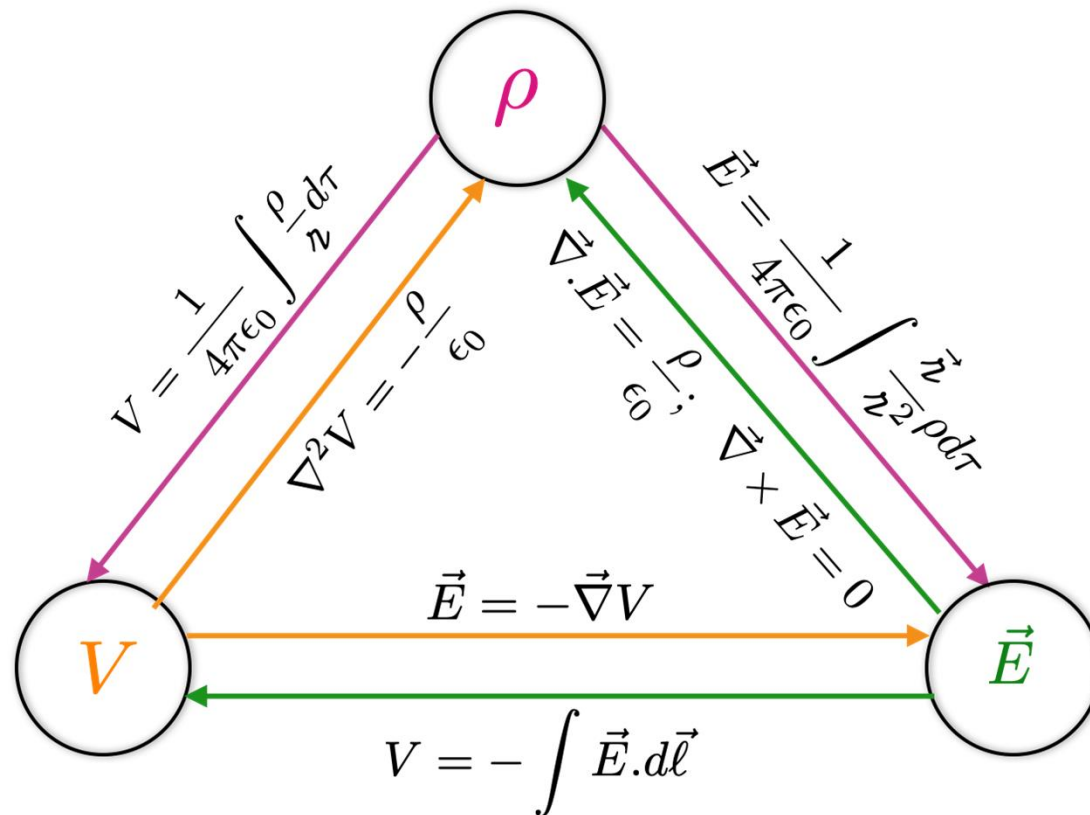
❖ Gauss's law

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \quad (\text{Integral form})$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{Differential form}) \quad \text{First Maxwell's Equation}$$

Important parameters in electrostatics

E , V and ρ are important parameters in electrostatics.



Electrostatic boundary conditions

The normal component of E is discontinuous by an amount σ/ϵ_0 at any boundary.

$$E_{above}^{\perp} - E_{below}^{\perp} = \frac{\sigma}{\epsilon_0}$$

The parallel component of E is continuous at any boundary.

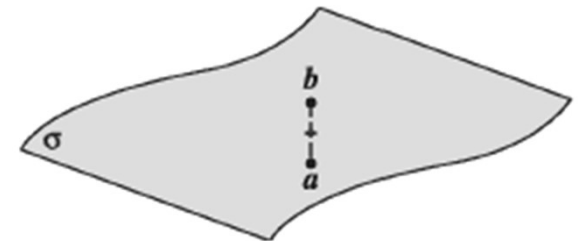
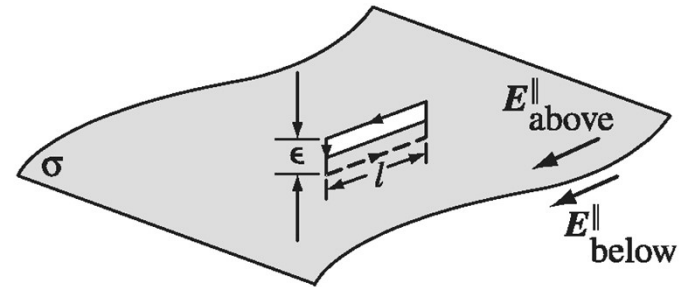
$$E_{above}^{\parallel} = E_{below}^{\parallel}$$

The potential is continuous at any boundary.

$$V_{above} = V_{below}$$

The gradient of the potential is discontinuous by an amount $-\sigma/\epsilon_0$ at any boundary.

$$\left(\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} \right) = -\frac{\sigma}{\epsilon_0}$$



Poisson's equation and Laplace's equation

The electric field is, $\mathbf{E} = -\nabla V$.

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \quad (\text{Poisson's equation})$$

In regions where there is no charge,

$$\nabla^2 V = 0 \quad (\text{Laplace's equation})$$

Uniqueness theorem

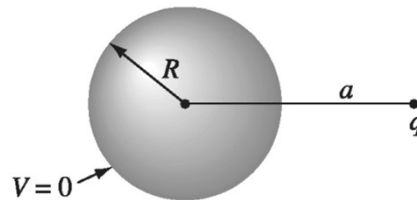
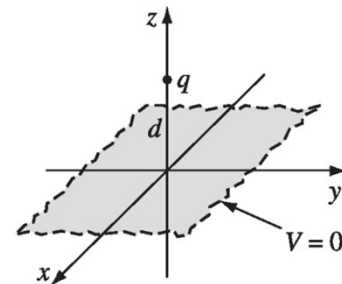
- Any solution of Laplace's equation which satisfies the same boundary conditions must be the only solution regardless of the method used. This is known as the *uniqueness theorem*.
- The theorem applies to any solution of Poisson's or Laplace's equation in a given region or closed surface.

Special techniques

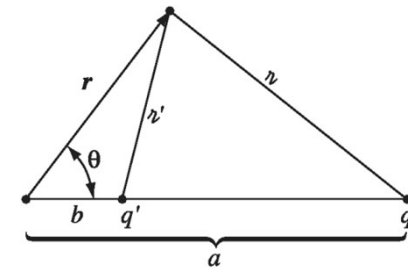
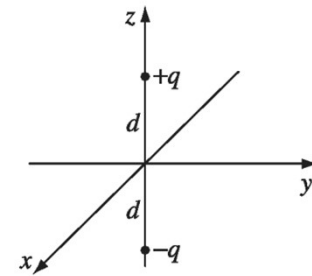
➤ Method of images:

Uniqueness theorem .

Original problem



Equivalent problem



➤ Method of separation of variables: $V(x, y, z) = X(x)Y(y)Z(z)$.

$$V(x, y) = (Ae^{kx} + Be^{-kx})(C \sin ky + D \cos ky)$$

$$C_n = \frac{a}{2} \int_0^a V_o(y) \sin \left(\frac{n' \pi y}{a} \right) dy$$

Electrostatics in matter

- Polarization results in the dipole moment, $\mathbf{p} = \alpha \mathbf{E}$.
- Torque on a permanent dipole in uniform field is $\mathbf{N} = \mathbf{p} \times \mathbf{E}$
- Force on a permanent dipole in a non-uniform field is $\mathbf{F} = q\Delta\mathbf{E} = (\mathbf{p} \cdot \nabla)\mathbf{E}$
- Potential at points far from the dipole is $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$
- The electric field due to a dipole is $\mathbf{E}_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$
- The bound surface charge density is $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$
- The bound volume charge density $\rho_b = -\nabla \cdot \mathbf{P}$

The electric displacement

- Any charge that is *not* a result of polarization is a free charge.
- In the presence of a dielectric, the Gauss's law is $\nabla \cdot \mathbf{D} = \rho_f$
- The curl of $\mathbf{D}$ is not always zero and so it cannot be expressed as the gradient of a scalar.
- The normal component of $\mathbf{D}$ is discontinuous by an amount σ_f at any boundary.

$$D_{above}^{\perp} - D_{below}^{\perp} = \sigma_f$$

- The parallel component of $\mathbf{D}$ is discontinuous at any boundary.

$$D_{above}^{\parallel} - D_{below}^{\parallel} = P_{above}^{\parallel} - P_{below}^{\parallel}$$

- For a linear dielectric medium,

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

Boundary conditions

- Any charge that is *not* a result of polarization is a free charge.
- In the presence of a dielectric, the Gauss's law is $\nabla \cdot \mathbf{D} = \rho_f$
- The curl of $\mathbf{D}$ is not always zero and so it cannot be expressed as the gradient of a scalar.
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$$D_{above}^{\perp} - D_{below}^{\perp} = \sigma_f$$

- The parallel component of $\mathbf{D}$ is discontinuous at any boundary.

$$D_{above}^{\parallel} - D_{below}^{\parallel} = P_{above}^{\parallel} - P_{below}^{\parallel}$$

- For a linear dielectric medium,

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

Fields in a dielectric medium

- **D** is related to the vacuum field by the relation, $\mathbf{D} = \epsilon_0 \mathbf{E}_{vac}$

- **E** in a medium is related to the vacuum field $\mathbf{E}_{vac}$ by the relation,

$$\mathbf{E} = \frac{1}{\epsilon_r} \mathbf{E}_{vac}$$

- Boundary conditions in terms of free charge on a medium are

$$\epsilon_{above} E_{above}^{\perp} - \epsilon_{below} E_{below}^{\perp} = \sigma_f$$

$$\epsilon_{above} \frac{\partial V_{above}}{\partial n} - \epsilon_{below} \frac{\partial V_{below}}{\partial n} = -\sigma_f$$

$$V_{above} = V_{below}$$

- The energy of the charge configuration in a medium is

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

- The potential energy of a dipole in an electric field is $U = -\mathbf{p} \cdot \mathbf{E}$

ANNOUNCEMENT

Mid-semester exam

Date: 28.02.2024 (Wednesday)

Time duration: 2 hours

Venue: As per the academic section allotment

Syllabus: Lecture 01 to 14

Mode: Subjective type (solve and write all steps in the question-cum-answer booklet)

CALCULATOR NOT ALLOWED

Questions